IGCSE 0607 Practice Questions — E4.1

Coordinates

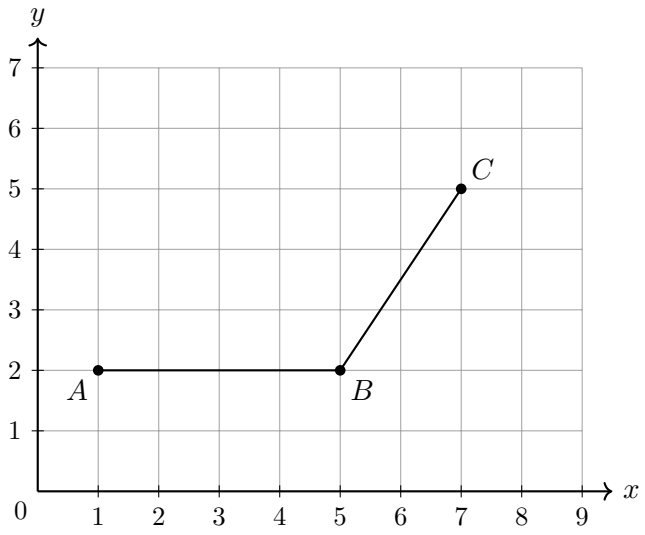
Total Questions: 12 — **Total Marks:** 81 Marks

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1 The diagram shows three vertices, $A(1, 2)$, $B(5, 2)$ and $C(7, 5)$, of a parallelogram $ABCD$.



(a) Find the coordinates of vertex D.

((..... , ...)) **[2]**

(b) Calculate the area of the parallelogram $ABCD$.

..... **[2]**

2 Points $P(-4, 2)$ and $Q(6, 2)$ are plotted on a Cartesian grid.

(a) Write down the coordinates of the midpoint of the line segment PQ.

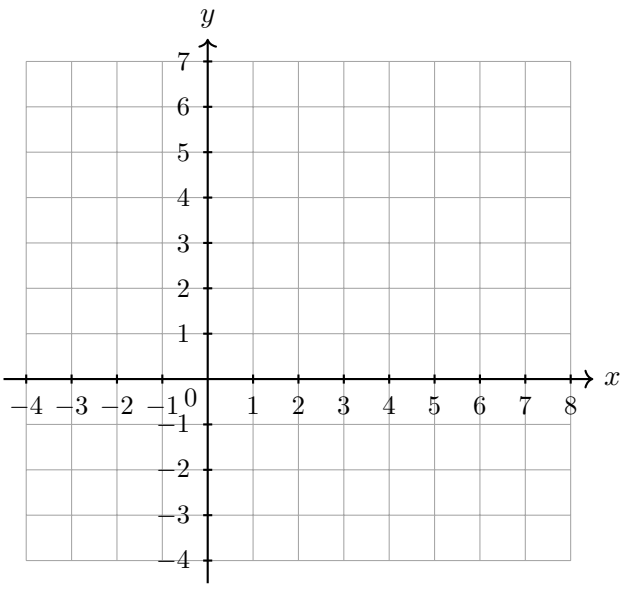
(..... , ...) **[1]**

(b) Point R lies on the positive y-axis.
The area of triangle PQR is 35 square units.
Find the coordinates of R.

(..... , ...) **[3]**

3 The points $A(-1, 3)$, $B(2, -1)$ and $C(6, 1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

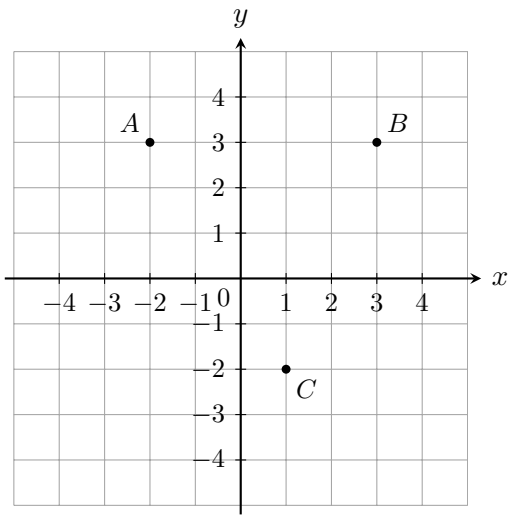
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

4 The diagram shows three points, A, B, and C, plotted on a Cartesian coordinate grid.



(a) Write down the coordinates of point A and point C.

$A =$ ..... , $C =$ ..... **[2]**

(b) Write down the coordinates of point D such that $ABCD$ is a parallelogram.

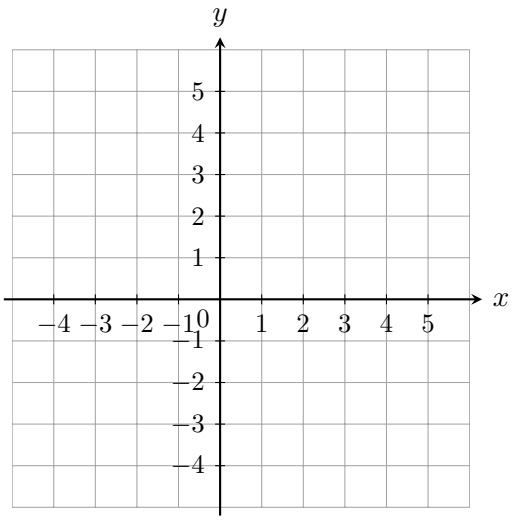
$D =$ (..... , ...) **[2]**

(c) Calculate the area of parallelogram $ABCD$.

..... square units **[2]**

5 In a Cartesian coordinate plane, a kite $PQRS$ has vertices $P(1, 5)$, $Q(4, 2)$, $R(1, -4)$, and $S(-2, 2)$.

(a) Plot the points $P(1, 5)$, $Q(4, 2)$, $R(1, -4)$, and $S(-2, 2)$ on the grid below and join them in order to form the kite $PQRS$.



[2]

(b) Write down the coordinates of the point where the diagonals of kite $PQRS$ intersect.

(..... , ...) **[1]**

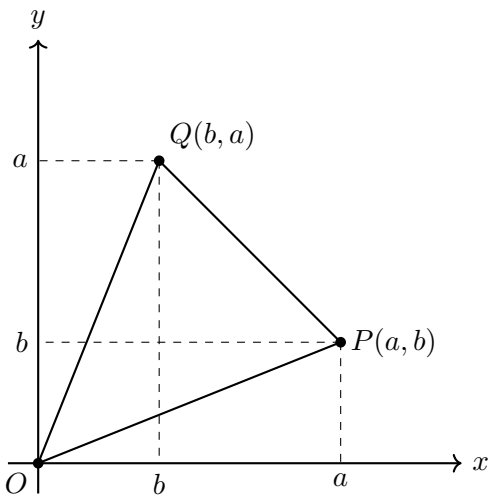
(c) Calculate the area of kite $PQRS$.

..... square units **[2]**

(d) Point P is reflected in the x-axis to give point P', and point Q is reflected in the y-axis to give point Q'.
Write down the coordinates of P' and Q'.

$P' =$ ..... , $Q' =$ ..... **[2]**

6 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

7 A rectangle has its sides parallel to the coordinate axes.
Two opposite vertices of the rectangle are $A(-2, -4)$ and $C(x, y)$, where $x > 0$ and $y > 0$.
The area of the rectangle is 54 square units and its perimeter is 30 units.

(a) Show that $x + y = 9$.

[2]

(b) Form an equation in x and solve it to find the two possible pairs of coordinates for C.

[4]

(c) For the case where $x > y$, write down the coordinates of the other two vertices of the rectangle.

[2]

8 Points A and B have coordinates $(3k, -k)$ and $(k+4, -5k)$ respectively, where k is a constant.

(a) The midpoint of the line segment AB lies on the y-axis.
Find the value of k.

$k =$ ..... **[2]**

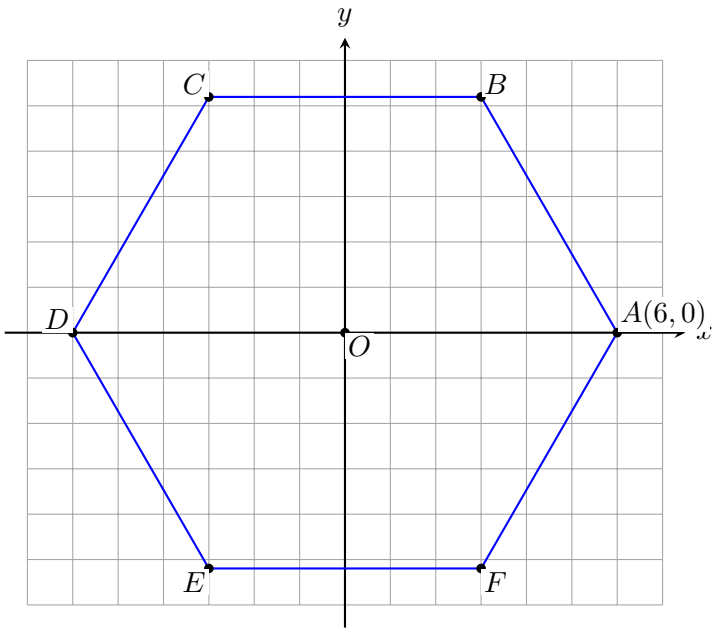
(b) Using your value of k from part (a), find the exact length of AB.
Give your answer in the form $a\sqrt{b}$ where b is a prime number.

..... **[3]**

(c) A point C lies on the x-axis.
The distance AC is equal to the distance BC.
Find the coordinates of C.

(..... , ...) **[4]**

9 A regular hexagon $ABCDEF$ has its center at the origin $O(0, 0)$.
The point A has coordinates $(6, 0)$.



(a) Write down the coordinates of point D.

(..... , ...) **[1]**

(b) Calculate the coordinates of point B.
Give your values correct to 2 decimal places.

(..... , ...) **[3]**

(c) Point P lies on the line segment AD such that $AP : PD = 1 : 3$.
Find the coordinates of P.

(..... , ...) **[2]**

10 A straight line has the equation $2y + 3x = 12$.
The line intersects the y-axis at A and the x-axis at B.
The point O is the origin $(0, 0)$.

(a) Find the coordinates of A and B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Point C is such that $OACB$ is a rectangle.
Write down the coordinates of C.

((..... , ...)) **[1]**

(c) Point P lies on the line segment AB and has an x-coordinate of 2.
Find the coordinates of P.

((..... , ...)) **[2]**

(d) Find the area of triangle OPB.

..... **[2]**

11 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]

12 A rectangle $ABCD$ has vertices $A(-3,-1)$, $B(5,-1)$, $C(5,k)$, and $D(h,k)$, where $k>0$.

(a) State the value of h and the quadrant in which point D lies.

$h =$ ..... , Quadrant: ..... **[2]**

(b) Given that the area of rectangle $ABCD$ is 52 square units, calculate the value of k.

$k =$ ..... **[3]**

(c) Point M is the centre of rectangle $ABCD$.
Find the coordinates of M.

$M =$ (..... , ...) **[2]**

(d) Point $E(p,q)$ lies on the y-axis such that triangle ABE has an area of 18 square units and $q>0$.
Find the coordinates of E.

$E =$ (..... , ...) **[3]**